LESSON PLAN

AIRCRAFT WIRING PRACTICES (ELECTRICAL WIRING AND INTERFACE SYSTEM) (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to helicopter, difference between fixed wing and rotary wing, history of helicopter, development, different rotor systems
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to helicopter, difference between fixed wing and rotary wing, history of helicopter, development, different rotor systems.
5. Specific Objectives	12.1.1.1 Trainee will be able to explain helicopter, difference between fixed wing and rotary wing and history of helicopter.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Nil
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a helicopter?
(b) Can you name one major difference between fixed wing and rotary wing aircraft?
(c) Who is credited with the first successful powered helicopter flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic components of helicopter • Fixed wing vs rotary wing aerodynamic differences • Historical timeline of helicopter development	L & Ds	PPT, IB	15 min
(b)	• Evolution of rotor systems and control mechanisms • Key milestones in helicopter technology • Modern helicopter design trends	L & Ds	PPT, PDF	20 min
(c)	• Classification of different rotor systems • Impact of rotor design on flight performance • Future developments in rotor technology	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the fundamental difference in lift generation between fixed wing and rotary wing aircraft.
(b) Describe one significant historical milestone in helicopter development.
(c) How does rotor system design influence the operational capabilities of a helicopter?

12. Home Assignment and Follow Up	Research and list three modern helicopter models, noting their specific rotor system types and primary operational roles.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of shaft axis, axis rotation, and plane of rotation tip path plane, blade pitch, rotor thrust, relative air flow, coning angle, limit of RPM.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min